

Solutions

5 MARK QUESTIONS

Question 1

Five bees were racing to the same flower.

A got to the flower before B, but after C.

D got to the flower before E, but after B.

What was the finishing order from first to last?

Answer: CABDE

Question 2

If $a \uparrow b$ means a^b and $a \downarrow b$ means $\sqrt[b]{a}$ then the value of: $((2 \uparrow 6) \downarrow 3) \uparrow 2$

Answer: 16

Question 3

How many cubes with edges of length 2cm can be made from a rectangular block 6cm long, 4cm wide and 2cm high?

$$(6 \times 4 \times 2) \div (2 \times 2 \times 2) = 6$$

Answer: 6 cubes

Question 4

The value of $2222^2 - 2221^2$ is...

$$\begin{aligned} & (2222 - 2221)(2222 + 2221) \\ &= (1)(4443) \\ &= 4443 \end{aligned}$$

Answer: 4443

Question 5

How many two-digit numbers, less than 50, are exactly 4 times the sum of their digits?

Answer: 4: {12;24;36;48}

Question 6

Increasing $800 - 10y$ by 10% gives $600 - 6y$.

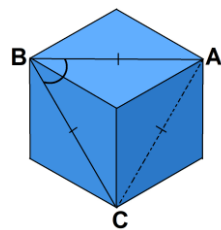
What is the value of y ?

$$\begin{aligned} \frac{11}{10}(800 - 10y) &= 600 - 6y \\ 880 - 11y &= 600 - 6y \\ 880 - 600 &= 11y - 6y \\ 280 &= 5y \\ 56 &= y \end{aligned}$$

Answer: 56

Question 7

In the cube, how big is angle $\widehat{ABC}$?



equilateral triangle so **Answer: 60°**

Question 8

You are allowed to perform **TWO** operations with numbers: **doubling** and **increasing by 1**. If you start from zero, what is the smallest number of operations needed to build up the number 100.

$$0+1=1$$

$$1 \times 2 = 2$$

$$2+1=3$$

$$3 \times 2 = 6$$

$$6 \times 2 = 12$$

$$12 \times 2 = 24$$

$$24+1=25$$

$$25 \times 2 = 50$$

$$50 \times 2 = 100$$

Answer: 9

Question 9

Three normal dice are rolled. What is the probability that they all show the same number?

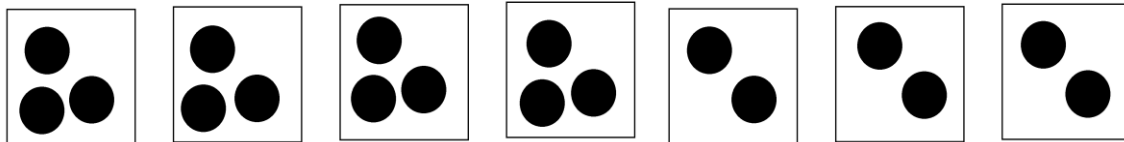
Answer: $\frac{6}{216}$ or $\frac{1}{36}$ or $0,02\bar{7}$

Question 10

25 pots of honey are distributed in 14 boxes. Each box contains 1, 2 or 3 honey pots. 7 boxes contain just one honey pot each. How many boxes will have only 2 honey pots in each?

Boxes left for 2 or 3 honey pots: $14 - 7 = 7$

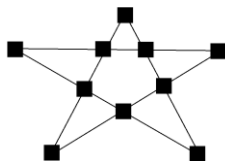
Honey pots left $25 - 7 = 18$



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10 MARK QUESTIONS**Question 11**

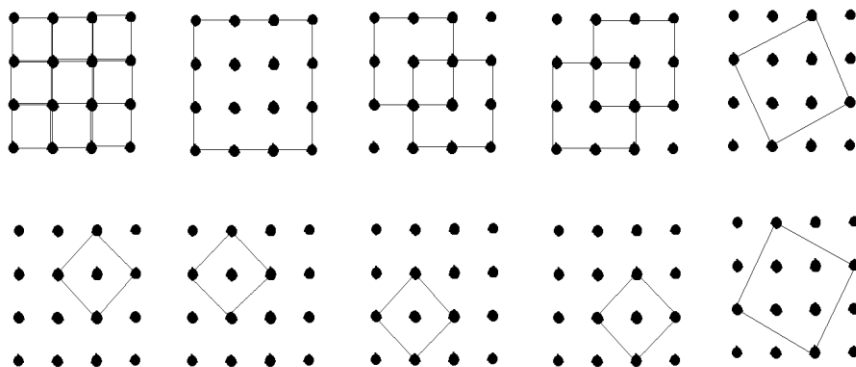
Layla has 10 bee hives. She arranges the hives in a shape of a _____ in order to have 5 rows with 4 hives in each row.



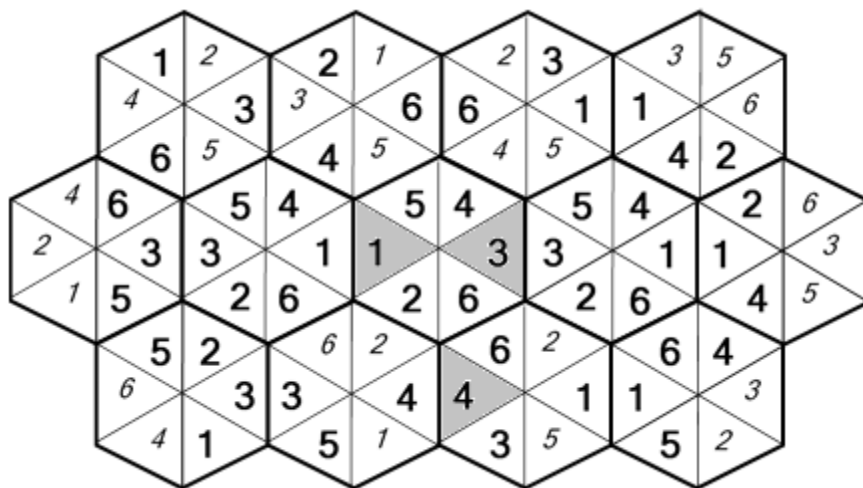
Answer: Star

Question 12

By joining dots together, how many squares can be drawn on a grid measuring 4 dots by 4 dots? Some examples have been provided below:



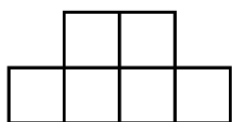
Answer: 20

Question 13

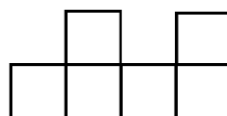
Answer: 8

Question 14

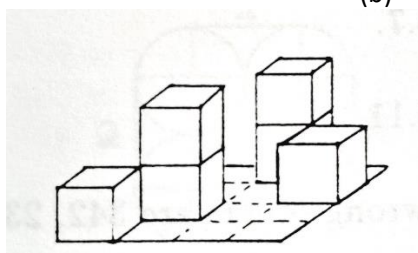
What is the smallest number of cubes needed to make the arrangement shown below, viewed (a) from the front and (b) from the side?



(a)



(b)



Answer: 6 cubes

Question 15

Use the clues to crack the code.

2	6	5	←	One number is correct and well placed
3	8	7	←	Nothing is correct
2	3	4	←	One number is correct but wrongly
4	7	1	←	Two numbers is correct but wrongly
?	?	?		

Answer 145

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Question 16

The natural numbers are arranged in the pattern below. In which row does 2022 lie?

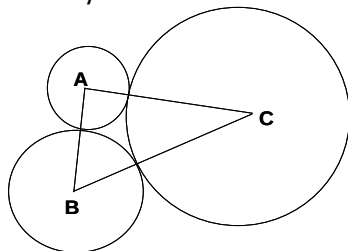
Row 1			4		12			20		
Row 2		2		7	10		15		18	23
Row 3	1		5	9	13		17		21	25
Row 4		3		8	11		16		19	24
Row 5			6		14				22	

Answer: Row 5

Question 17

The vertices of a 3-4-5 right triangle are the centres of three mutually externally tangent circles as shown. $AB = 3$; $AC = 4$; $BC = 5$. What is the sum of the areas of these circles?

Leave your answer in terms of π .



$$r_a + r_b = 3$$

$$r_a + r_c = 4$$

$$r_b + r_c = 5$$

Solving simultaneously $r_c = 3$

$$A = \pi r_a^2 + \pi r_b^2 + \pi r_c^2$$

$$A = \pi(4 - r_c)^2 + \pi(5 - r_c)^2 + \pi(r_c)^2$$

$$A = \pi(4 - 3)^2 + \pi(5 - 3)^2 + \pi(3)^2$$

$$A = 14\pi$$

Answer: 14π

Question 18

What is the angle between the minute hand and hour hand at quarter past three?

Answer: $7,5^\circ$

Question 19

Stranded on a deserted island, Sam and his crew are left with only a 40 litre container of milk. To conserve the milk they decide that on the first day they will drink one litre of milk and then refill the container back up with water. On the 2nd day they will drink 2 litres and refill the container. On the 3rd day they will drink 3 litres and so on... By the time all the milk is finished, how much water have they drunk altogether?

Solution

It is given that they have a 40 litres container of milk. Also, they will drink 1 litre on the first day and refill the container with water, then drink 2 litres on the second day and refill the container, then drink 3 litres on the third day and refill the container, and so on till 40th day. Thus at the end of 40 days, they must have drunk $(1 + 2 + 3 + 4 + \dots + 38 + 39 + 40) = 820$ litres of liquid.

Out of those 820 litres, 40 litres is the milk which was in the container initially. Hence, they must have drunk 780 litres of water.

Answer: 780 litres

Question 20

A faulty car odometer proceeds from digit 3 to digit 5, always skipping the digit 4, regardless of position. For example, the odometer changes immediately from 000039 to 000050. If the odometer now reads 002022, how many kilometers has the car actually travelled?

From 0 – 100 there are 19 numbers with the digit 4

From 400 – 500 there are 100 numbers with the digit 4

So, from 0 – 1000 there are $19 \times 9 + 100$ numbers with the digit 4

So, from 0 – 2000 there are $2(19 \times 9 + 100)$ numbers with the digit 4

From 2000 to 2022 there are 2 numbers with the digit 4

So, from 0 – 2022 there are $2(19 \times 9 + 100) + 2 = 544$ numbers with the digit 4

So the car has travelled $2022 - 544 = 1478$ km

Answer: 1478 km